

Freeze, Validate, Report: Auditing Urban Station Plans with Common Evidence

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Abstract

Transport authorities comparing urban station plans need to know whether score differences reflect the plans or their evaluation inputs. If spatial units, included trips, or candidate blocks are reconstructed from each plan's access outcomes, apparently comparable scores can refer to different evidence bases. *Freeze-Validate-Report* fixes an evaluation contract before comparison: target trips, proxy groups, reporting cells, pools of candidate blocks, and a pre-declared lexicographic selection rule. A two-point example shows that plan-specific observation selection can reverse the target mean ordering. We compare seven station-plan generators spanning classical clustering and facility location, fairness-oriented adaptations, and a grid control. We score every plan on common validation evidence using endpoint distance, the sum of the distances from a trip's origin and destination to their respective nearest stations; a five-metric rule selects one for a single test report. For Porto, the rule selects IFkCO, our adaptation of individually fair k -center with outliers; its test overall and worst-group 90th percentiles (p90) are 805 m and 872 m. For Chicago, the rule selects Grid, a grid-based control; the corresponding values in a later test window on the same day are 1,404 m and 1,656 m. Two post hoc checks show why the contract records its inputs: expanding the candidate set from 300 to 600 changes the selection in both cities, while a Chicago design that holds the hour fixed across three dates selects Priority, a priority-center adaptation. A secondary proportional mean fairlet (PMF) diagnostic shows that plan-specific proposal pools can change diagnostic ranks; admissibility and solver bounds apply only to the sampled pool. The package supplies code, hashes, decision logs, and proposal pools. We establish auditability within a declared contract, not stable performance after deployment; proxy definitions and metric priorities remain the authority's choices.

CCS Concepts

• **Computing methodologies** → **Cluster analysis**; • **Information systems** → *Spatial-temporal systems*; • **Theory of computation** → *Facility location and clustering*; • **General and reference** → *Evaluation*.

Keywords

urban mobility, facility location, spatial accessibility, fair clustering, comparative evaluation, urban decision-making

1 Introduction

Consider a transport authority choosing among proposals for k mobility stations. A station plan is a set of k candidate sites. The authority wants to compare overall access and access across area-level proxy groups. For an observed trip, endpoint distance sums the origin-to-nearest-station and destination-to-nearest-station distances. The comparison is meaningful only when every plan is scored on the same trips and reporting cells under the same selection rule. If those inputs are reconstructed from each plan's own outcomes, the authority cannot tell whether a favorable score reflects better access or a more favorable evidence base. Section 3.1 gives an example with two demand points in which such reconstruction reverses the ranking.

Station placement distributes access burdens across neighborhoods. Classical facility location objectives summarize this burden through a demand-weighted total or a maximum distance [9, 23]. Other methods instead optimize the largest burden among groups or a priority objective [2, 8, 14]. A planning audit may also group trips into spatial reporting units and ask whether fixed proxy groups receive comparable outcomes.

Exact optimization does not resolve this evaluation problem: an exact method only optimizes the supplied instance. Without a separately proved invariance condition, comparable scores require a common instance. The post hoc checks in Section 6.3 illustrate sensitivity to these inputs: changing the candidate set or temporal window changes the selected plan. The contract records this dependence; whether its fairness definition is appropriate remains a separate normative question.

Throughout, the planning authority declares the comparison and chooses; the analyst applies the declared inputs; and the auditor checks the record. The *evaluation contract* comprises all inputs declared before a comparison. *Common evidence* denotes the observations, reporting partition, and pool of candidate blocks shared across plans. The *selection rule* denotes the ordered validation metrics used to choose a plan.

We study the following question:

Research question. Can a planning authority compare station plans on a common evidence base while keeping evidence construction, validation-based selection, and final test reporting traceable and separate?

Freeze-Validate-Report has three phases. First, the authority records its choices; training data determine candidate sites, plans, and the coordinate frame. Second, the analyst scores all plans on common

validation evidence, and the authority’s rule chooses one; the Pareto front is descriptive only. Third, the program hashes the selected plan and decision record before loading test data, then evaluates only that plan. Test outcomes cannot alter the choice.

We evaluate two public origin–destination (OD) settings. Porto taxi trips use four census socioeconomic status (SES) strata (Porto-SES). Chicago ride-hail trips use four Hardship Index strata defined from community areas (Chicago-Hardship). In both settings, we treat observed trip endpoints as demand for hypothetical stations: the datasets contain observed OD trips, not trips made between the hypothetical stations. Section 5 gives the sources, periods, and sampling procedure. The seven generators span classical clustering and facility location, fairness-oriented adaptations, and a grid control. Together, they let us study comparative evaluation for urban decision intelligence while holding the decision target fixed. Learned or hybrid generators can enter as additional plan sources under the same contract.

The contributions are:

- a three-phase evaluation protocol that freezes the comparison contract, selects one plan on validation data, and reports it once on test data;
- an audit-consistency condition that identifies when plan scores refer to the same observations and comparison inputs, together with a minimal rank-reversal example;
- a two-city comparative evaluation of seven station-plan generators spanning clustering, facility location, fairness-oriented adaptations, and a grid control, using every sampled trip, a grid and validity thresholds fixed before validation, and a predeclared lexicographic selection rule; and
- a secondary proportional mean fairlet (PMF) ablation of common versus plan-specific proposal pools, post hoc design and bootstrap sensitivity checks, and a reproducibility package with phase logs, hashes, proposal pools, and separately implemented verification code.

These retrospective demonstrations were neither used nor validated by the Porto or Chicago authorities and do not estimate how performance or rankings vary after deployment.

2 Related Work and Positioning

Urban AI systems produce predictions, spatial representations, and infrastructure analyses that can inform planning decisions [15, 18, 19, 22]. We study the comparison layer of urban decision intelligence: when classical, learned, or hybrid methods supply alternative plans, a planning authority needs scores that refer to a common target and common evidence.

Model cards report intended uses, performance across groups, and evaluation procedures for trained models [16]. End-to-end internal audit frameworks call for traceable records across development stages [20]. Our condition is narrower: it formalizes when alternative plans are scored against the same observations and evaluation inputs.

Classical k -center and p -median objectives motivate our geometric and demand-weighted baselines [9, 23]. Jung et al. define individual service relative to a local neighborhood radius [14]; priority-center work likewise scales service distances by a radius defined

for each point and accommodates outliers [2]. Socially fair clustering controls the largest group cost [8], while individually fair k -center with outliers (IFkCO) combines a local service requirement with explicit exclusions [10]. We adapt these ideas to choose stations from a finite set of sites. We compare the resulting plans, not the original algorithm families. The audit includes every validation and test trip even when a training objective allows outliers.

Fairlets are small blocks with a prescribed group composition that can then be clustered as indivisible units [3]; fair clustering with outliers motivates explicit accounting for who is left out [1]. Our secondary blocks impose exact counts for each proxy group, a physical radius, and a limit on differences in mean station access among groups. We use these blocks only for diagnosis. Hierarchy construction and fairlet approximation guarantees lie outside the study.

Spatial equity results can also change with the scale and zoning of reporting units. This dependence is known as the modifiable areal unit problem (MAUP). Javanmard et al. demonstrate such sensitivity for public-transit reliability equity [13]. We fix one grid to prevent each plan from choosing its own units; the choice of scale is subject to MAUP.

Work on adaptive data analysis separates exploratory reuse from valid holdout inference [6]. Our narrower concern is that an analyst must first score two plans against the same stated target. Roberts et al. recommend blocked validation when observations have temporal or spatial dependence [21]. Our primary splits preserve temporal order but use only one sequence; blocked or rolling-origin validation would require additional windows.

Work on measurement models emphasizes that an operational proxy may differ from its intended construct [12]. Likewise, transport accessibility includes land use, opportunities, time, and individual constraints beyond a distance summary [7]. These partial measures must therefore be declared as decision inputs rather than treated as neutral descriptions of a city.

3 Problem and Evaluation Contract

We use $\mathcal{D}$ for an evaluation dataset and A for an audit implementation.

3.1 Common evidence and rank reversal

An earlier PMF constructor in our own pipeline used each plan’s origin-access outcomes to assign trips to block anchors, the seed points from which blocks are grown. It therefore supplied a different proposal pool to each plan’s diagnostic optimization, even with trips and constraints held fixed. Section 5.3 describes this concrete instance of plan-dependent evidence; the following simplified example isolates the effect of changing the observations being scored.

Example with two demand points. Suppose a planning authority declares mean access burden across two neighborhoods, represented by one demand point each. Plan S_1 yields 0 m and 1,000 m, with mean 500 m; plan S_2 yields 400 m and 500 m, with mean 450 m. The declared target therefore ranks S_2 first. An analyst instead retains each plan’s best-served point and reports 0 m for S_1 and 400 m for S_2 . The authority would select S_1 although its target ranks S_2 first. Each value is correct for the retained point but is a minimum, not the declared mean. The reversal occurs because the evidence

depends on the plan; its direction depends on the metric and outcomes.

Let $\mathcal{D} = (x_i, g_i)_{i=1}^n$ be a fixed evaluation dataset, where $x_i = (o_i, d_i)$ contains trip i 's origin and destination, g_i is its proxy group label, and $[n] = \{1, \dots, n\}$. Let $a_i(S)$ be its combined endpoint distance under plan S , as defined in Equation 5 below, and let $y(S) = (a_i(S))_{i=1}^n$ be the resulting outcome vector. Before comparison, the study fixes the target evidence

$$\mathcal{E}^* = (J^*, \mathcal{P}^*, C^*) \quad \text{and} \quad \theta(S) = F(y(S); \mathcal{D}, \mathcal{E}^*), \quad (1)$$

where J^* is the set of target observations, $\mathcal{P}^*$ is a reporting partition, and C^* is an optional pool of candidate blocks. An implemented audit A may instead supply

$$\mathcal{E}_A(S) = (J_A(S), \mathcal{P}_A(S), C_A(S)), \quad (2)$$

where the three components again specify observations, a partition, and a pool of candidate blocks. A deterministic score function F returns

$$q_A(S) = F(y(S); \mathcal{D}, \mathcal{E}_A(S)). \quad (3)$$

Components unused by a score are empty. We say that the implementation conducts an audit *with common evidence for the stated target* when $\mathcal{E}_A(S) = \mathcal{E}^*$ for every compared plan. The analyst may use geometry and fixed proxy labels to construct $\mathcal{E}^*$, but not access outcomes from individual plans.

Audit-consistency condition (by definition). For fixed $\mathcal{D}$, F , and target evidence $\mathcal{E}^*$, an implementation with common evidence satisfies $q_A(S) = \theta(S)$ for every plan and therefore

$$q_A(S) - q_A(S') = F(y(S); \mathcal{D}, \mathcal{E}^*) - F(y(S'); \mathcal{D}, \mathcal{E}^*). \quad (4)$$

The equality follows directly by substituting $\mathcal{E}_A(S) = \mathcal{E}_A(S') = \mathcal{E}^*$ into the two score definitions, so only the outcome vector depends on the plan. Without common evidence or a separately proved invariance condition, scores computed from plan-dependent evidence may fail to reproduce the target ordering. The example with two demand points uses the same function F to average the retained observations, but the retained set depends on the plan. Its two reported values rank S_1 ahead of S_2 and do not recover the target ordering, which ranks S_2 ahead of S_1 .

Satisfying this condition ensures only that both scores represent the same target and that their difference can be attributed to the outcome vectors. Causal or normative fairness claims require additional assumptions. In the primary audit, we fix $J^* = [n]$ and one partition. Section 4.2 applies the same condition to the secondary diagnostic.

3.2 OD access and fixed proxy groups

Let an evaluation split contain n trips. Trip i has origin $o_i \in \mathbb{R}^2$, destination $d_i \in \mathbb{R}^2$, and proxy group label $g_i \in G$ assigned from its origin area. The finite set G contains all proxy groups, and U denotes the finite set of candidate sites. A station plan $S \subseteq U$ has budget $|S| = k$. Its endpoint distance is

$$a_i(S) = \min_{s \in S} \|o_i - s\|_2 + \min_{s \in S} \|d_i - s\|_2. \quad (5)$$

It sums origin access and destination egress but excludes travel between stations; smaller values are better. The evaluation artifact (the saved outputs of each run) reports the overall mean, median, 90th percentile (p90), 95th percentile (p95), and maximum, as well

as these summaries separately for origin and destination access. Let $I_g = \{i : g_i = g\}$ contain the $n_g = |I_g|$ trips in group g . Write $Q_q(X)$ for the q -quantile of a finite multiset X and $\bar{a}_g(S) = \frac{1}{n_g} \cdot \sum_{i \in I_g} a_i(S)$ for group g 's mean endpoint distance. The two group metrics are

$$W_{90}(S) = \max_{g \in G} Q_{0.9}(\{a_i(S) : i \in I_g\}), \quad (6)$$

$$D_G(S) = \max_g \bar{a}_g(S) - \min_g \bar{a}_g(S). \quad (7)$$

The worst-group p90, W_{90} , is the largest p90 among the groups, while D_G is the difference between the highest and lowest group mean. They summarize tails and means across the fixed proxy groups. Individual fairness would require a separate criterion for each trip.

3.3 Fixed grid constructed without plan outcomes

We evaluate every plan on the same partition $\mathcal{P}^*$. We place a square grid in the coordinate frame fitted to training geometry and fix its cell size and origin before validation. We assign each validation or test trip to a cell using only its origin coordinates.

A cell is *auditable* when it contains at least $m_{\min}$ trips. Let $\mathcal{P}_{\text{aud}}$ be the auditable cells and let $\bar{a}_p(S)$ be the mean endpoint distance in cell p . The artifact reports both the full range of cell means and the cell gap

$$D_{\text{cell}}(S) = Q_{0.9}(\{\bar{a}_p(S) : p \in \mathcal{P}_{\text{aud}}\}) - Q_{0.1}(\{\bar{a}_p(S) : p \in \mathcal{P}_{\text{aud}}\}). \quad (8)$$

The 90th–10th percentile difference reduces, but does not remove, the influence of extreme cells. The audit is valid only if at least two cells are auditable and, together, contain a predeclared minimum share of all trips. Cell membership is therefore invariant across plans. The artifact also reports differences between groups within a cell when every group has enough observations; these sparse within-cell diagnostics are descriptive only and do not enter the selection rule. The results foreground the five selection metrics in Equation 10 and additionally report the trip share in auditable cells for the selected plans.

For a fixed limit L on endpoint distance, we further report

$$\alpha_L(S) = \frac{1}{n} \cdot \sum_{i=1}^n \mathbf{1}[a_i(S) \leq L]. \quad (9)$$

Here $\mathbf{1}[\cdot]$ equals one when its condition holds and zero otherwise. $\alpha_L(S)$ is therefore the share of trips whose endpoint distance is at most L . All n sampled trips are retained in the primary audit in Equations 5–9; cell summaries use the auditable cells defined above. Section 5 separately reports losses during data sourcing and preprocessing.

3.4 Selection rule fixed before validation

We use the seven generators to build the plan portfolio from training data. Because cell support is independent of the station plan, the audit-validity condition in Section 3.3 applies to the entire split rather than to individual plans. For a valid split, the predeclared lexicographic selection rule minimizes the vector

$$(W_{90}(S), Q_{0.9}(\{a_i(S)\}_{i=1}^n), D_{\text{cell}}(S), -\alpha_L(S), D_G(S)). \quad (10)$$

The rule first minimizes the worst-group p90. It then breaks ties, in order, by overall p90, cell gap, share within L , and group mean gap. To describe trade-offs, we also identify the validation Pareto front: no other compared plan is no worse in all five components and better in at least one. With the same five components, a lexicographic minimizer is necessarily nondominated, so restricting the implementation to that front leaves the choice unchanged. Exact ties are broken by the lexicographic order of the method identifier. This deterministic rule uses point estimates with no positive indifference threshold; it does not establish statistical or practical superiority. For prospective decisions, the contract should also specify how practically indistinguishable scores are handled, for example through predeclared indifference thresholds. We do not apply such thresholds retrospectively. Other planning decisions may require a different metric priority order. These five audit metrics determine selection; PMF is secondary.

After writing the selected plan S^* and the selection hash, the program loads the test data and evaluates S^* once. It does not compute test metrics for the other six plans; selection therefore precedes all test metrics.

4 Method

Algorithm 1 applies the authority-supplied contract and metric priorities from training through reporting.

4.1 Seven methods for generating station plans

The generators target spatial coverage, demand-weighted access, tail access by group, or individual priority. The audit described above evaluates all resulting plans on the same validation evidence. All generators receive the same training trips, 300 candidate sites drawn from endpoints observed in those trips, and budget $k = 200$. This modeling choice restricts siting to observed demand locations rather than independently authorized sites; it makes the task closer to demand clustering than general site selection. Its effect on the relative performance of Grid is not isolated. The primary configuration has a constrained candidate universe: the large $k/|U| = 2/3$ ratio limits portfolio diversity, with pairs sharing 113 to 198 stations in Porto and 128 to 198 in Chicago. The post hoc check with 600 sites provides a complementary comparison with greater plan diversity (Section 6.3). We compute all seven plans before constructing the PMF pool:

- (1) **Gonzalez** uses farthest-first traversal of U [9].
- (2) **K-medoids** is a bounded k -medoids swap heuristic over candidate-site demand. Starting from k -means++ medoids, it applies the best improving swap in each of at most two exhaustive rounds.
- (3) **Demand** applies a demand-weighted p -median objective and the same bounded swap search to all observed origin and destination demands.
- (4) **Group-fair mean** minimizes the highest proxy-group mean endpoint distance, then the overall mean, so the group with the highest average distance has priority [8].
- (5) **IFkCO** adapts the individually fair k -center with outliers objective to OD trips and applies a 10% training outlier allowance. For each pair of trips, it compares two endpoint matchings: origin to origin and destination to destination,

or the crossed matching. It chooses the matching with the smaller maximum endpoint distance, maps the representative endpoints to the nearest sites in U , and fills the remaining budget by farthest-first selection [10]. The later audit nevertheless includes every validation and test trip.

- (6) **Priority** applies a priority-center adaptation. It minimizes the worst ratio of a group's maximum endpoint distance to that group's p90 under the Gonzalez baseline plan, followed by the overall maximum and mean [2].
- (7) **Grid** is a control that places sites near the centroids of the grid cells containing the most trip endpoints and fills the budget by farthest-first. It uses endpoint counts and geometry alone.

Group-fair mean, IFkCO, and Priority are adaptations for station placement; Grid is a control.

4.2 Secondary proportional mean fairlet diagnostic

We use this secondary diagnostic to study how a plan-dependent proposal pool changes the basis of comparison; it does not select the station plan. A *fairlet* is a small mixed-group block that a later clustering step treats as one unit. The diagnostic has three stages. First, we construct one common pool of local blocks from trip origins and fixed proxy labels. Second, for each station plan, we identify blocks whose group mean origin-access values differ by at most ε_m . Third, the same set-packing model selects an admissible collection of disjoint eligible blocks. The pool is shared across plans; plan outcomes affect only block eligibility. We measure origin access only, so equal means can reflect leveling down as well as improvement.

We fit the group template $t = (t_g)_{g \in G}$ from training group counts; t_g is the required number of group- g trips in every block. Conceptually, $\mathcal{B}^*$ contains all blocks that meet the geometric and compositional requirements below. Because enumerating $\mathcal{B}^*$ is impractical, the implementation constructs a deterministic proposal pool $C \subset \mathcal{B}^*$ before inspecting any station outcome. Every block $B \in C$ has exact counts $|B \cap I_g| = t_g$ and a medoid radius at most $R_{\max}$. Here the medoid is the block member with the smallest total straight-line origin distance to the other members, and the radius is the largest distance from that medoid. We reuse the same proposal pool for all seven validation plans. After fixing the pool, each plan's outcomes determine which blocks meet the threshold. For a plan S , let $\bar{a}_g^{\text{orig}}(B, S)$ be the mean distance from an origin to its nearest station among group g trips in B , and let $C_S \subseteq C$ be the eligible subset. A block $B \in C$ belongs to C_S when

$$\max_{g, g' \in G} |\bar{a}_g^{\text{orig}}(B, S) - \bar{a}_{g'}^{\text{orig}}(B, S)| \leq \varepsilon_m. \quad (11)$$

The fixed threshold ε_m is measured in m; every plan uses the same value.

For each $B \in C_S$, binary variable z_B indicates whether the model selects that block. We solve the following set-packing model:

$$\max \quad M = \sum_{B \in C_S} z_B, \quad (12a)$$

$$\text{s.t.} \quad \sum_{B \in C_S: i \in B} z_B \leq 1 \quad \forall i, \quad (12b)$$

$$M \geq M_{\min}, \quad (12c)$$

$$\left| M \cdot \left(\frac{b}{n} - \frac{t_g}{n_g} \right) \right| \leq \delta_{\text{out}} \quad \forall g, \quad (12d)$$

$$z_B \in \{0, 1\} \quad \forall B \in C_S.$$

Here $z_B = 1$ means that block B is selected, and the first constraint prevents any trip from appearing in two selected blocks. We define $b = \sum_g t_g$; M is the number of selected blocks, and $M_{\min}$ is the smallest block count that satisfies both the overall inclusion floor and every group inclusion floor. With inclusion floor ρ_{in} , a fixed template gives overall inclusion $\frac{M \cdot b}{n}$ and inclusion $\frac{M \cdot t_g}{n_g}$ for group g , so $M_{\min} = \max(\lceil \rho_{\text{in}} n / b \rceil, \max_g \lceil \rho_{\text{in}} n_g / t_g \rceil)$. The exclusion-rate constraint limits the *exclusion-rate gap*, the largest absolute difference between overall and group-specific exclusion rates, to δ_{out} . Whenever $t_g/b \neq n_g/n$, the coefficient of M in group g 's exclusion-rate-gap constraint is nonzero, so that constraint can also impose an *upper* bound on M . Thus the inclusion floors, gap limit, and packing constraints define an interval of admissible block counts; $M \geq M_{\min}$ alone is insufficient. A result is admissible only when this interval contains the returned positive M .

SciPy/HiGHS solves the mixed-integer model in Equation 12 with a deterministic 10,000-node budget and zero requested relative gap. The incumbent is the best feasible packing found, the upper bound limits the size of any eligible packing in C_S , and their relative difference is the mixed-integer-programming (MIP) gap. We also report node count, completion status, and maximum selected radius. We use “optimal” only relative to the eligible subset induced by the generated pool C .

5 Experimental Setup

We instantiate the planning comparison retrospectively on two datasets with fixed time splits, selection criteria, and deterministic search budgets.

5.1 Temporal data and proxy groups

Porto-SES uses public taxi trajectories joined by origin location to four Census 2021 socioeconomic strata [11, 17]. Chicago-Hardship uses Transportation Network Provider trips joined by pickup community area to four Hardship Index strata [4, 5]. Both assignments describe a socioeconomic score for the trip's origin area. The resulting proxy labels contain no traveler attributes or verified residence data [12]. The Porto mobility records are from 2013 and the Census is from 2021; Chicago trips are from 2022 and the Hardship Index source covers 2008–2012.

For Porto, we retain trajectories in each period with at least ten recorded points and positive duration. After these filters, the three windows contain 9,815, 9,260, and 9,081 trips for train, validation, and test. We sample 5,000 with seeds 701, 708, and 722. We spatially join origin points to Census sections. When a point does not fall

Algorithm 1 Freeze–validate–report protocol. PMF denotes proportional mean fairlet.

Require: splits $D_{\text{tr}}, D_{\text{val}}, D_{\text{test}}$; station budget k ; predeclared audit settings

- 1: Fix the frame and U from D_{tr} ; record grid, groups, thresholds, and selection rule
- 2: Generate seven station plans from D_{tr} without PMF scores
- 3: Build one fairlet proposal pool on D_{val} without plan outcomes
- 4: **for** each plan S generated from D_{tr} **do**
- 5: Compute the five audit metrics on the fixed grid for D_{val}
- 6: Solve the fairlet packing on the same pool for diagnosis only
- 7: **end for**
- 8: Record the Pareto front; lexicographically select S^* ; hash the decision
- 9: Load D_{test} ; report the audit metrics for S^* exactly once
- 10: Build one test proposal pool without plan outcomes and report PMF for S^*
- 11: **return** selected plan, test audit, PMF result for the generated pool, manifests

inside a section, we assign it to the nearest section; this affects 2, 0, and 1 sampled points. For Chicago, we require finite pickup and drop-off centroids, duration, trip miles, and a Hardship score for the pickup community. Before sampling, 2,699, 4,762, and 3,395 rows are eligible; 0, 6, and 2 otherwise valid rows lack the score. Seeds 701, 801, and 1001 select 2,500 per role. The metadata do not enumerate every raw row rejected for malformed geometry or trip fields. Here, “zero exclusion” means that no sampled trip is excluded from the primary audit.

Each prepared split stores coordinates in a local equirectangular frame centered at its own mean origin, using Earth radius 6,371 km. Before evaluation, we transform validation and test coordinates into the training split's frame. All station distances and grid assignments therefore use one training frame. We divide each dataset's area-level scores into quartiles. A higher Porto label means higher constructed SES; a higher Chicago label means greater hardship. Tables present the source-index quartiles in increasing order (very low, low, middle, high). The quartiles are computed over areas, not sampled trips; this explains the unequal group counts across the fixed splits.

We fixed all periods before comparing plans. A gap separates validation from the later test window, which we used only after developing the protocol. The stored run sequence declares the selection decision before loading the test data. The primary Chicago train, validation, and test windows are all on June 1, 2022: 07:00–08:00, 08:00–09:00, and 10:00–11:00. Clock time is therefore confounded with split role in the primary design.

5.2 Selection and PMF parameters fixed before validation

Porto-SES uses grid cells of width 3 km and a limit $L = 2$ km on endpoint distance. Chicago-Hardship uses 5 km cells and $L = 5$ km. A cell needs 25 trips; at least two cells and 90% of trips must be auditable. Grid origins are $(0, 0)$ in each training coordinate frame.

Table 1: Primary time splits and sampled counts for each proxy group. All Chicago intervals are on June 1, 2022, and every interval includes its start but not its end. VL/L/M/H denote very low, low, middle, and high source-index quartiles.

Setting	Role	Period	n	group counts VL/L/M/H
Porto-SES	train	Jul 1–8, 2013	5,000	1387/525/432/2656
	validation	Jul 8–15, 2013	5,000	1423/521/494/2562
	test	Jul 22–29, 2013	5,000	1508/501/454/2537
Chicago-Hardship	train	Jun 1, 07:00–08:00	2,500	150/152/206/1992
	validation	Jun 1, 08:00–09:00	2,500	159/193/196/1952
	test	Jun 1, 10:00–11:00	2,500	127/126/191/2056

The set of candidate sites built from training data has $|U| = 300$, the station budget is $k = 200$, and bounded swap methods use at most two rounds. The cells group nearby trip origins for the spatial comparison; L determines the reported share of trips within the stated endpoint-distance limit.

We also fix the secondary PMF parameters. Template entries below use display order (very low, low, middle, high). Porto-SES uses block size 12, template (3, 1, 1, 7), $\epsilon_m = 250$ m, and $R_{\max} = 3$ km. Chicago-Hardship uses block size 15, template (1, 1, 1, 12), $\epsilon_m = 500$ m, and $R_{\max} = 5$ km. Both require inclusion floors of $\rho_{\text{in}} = 0.25$ overall and for every group, and $\delta_{\text{out}} = 0.30$. Pool construction uses seeds 101–105, three refill rounds, and six attempts per seed. We fix search effort by these attempt counts instead of wall-clock time.

5.3 Ablation with plans and data held fixed

We test whether using plan outcomes to construct the proposal pool changes the PMF diagnostic while holding the seven validation plans and trips fixed. The first design gives every plan the same stored pool, which uses only geometry and group labels. The second design reproduces an earlier constructor from our own pipeline. It builds a separate pool for each plan and uses origin access when assigning points to block anchors and when checking the difference between group means in each resulting block.

Both designs use the same templates, limits on radius and differences between group means, and requirements for inclusion and exclusion. They also use seeds 101–105, three refills, six attempts per seed, and at most 5,000 unique proposed blocks. We give the same binary set-packing model a 10,000-node limit.

The numbers of proposed blocks may differ because the two constructors define different pools. For display only, we apply a fixed diagnostic order. It first ranks admissible results ahead of unresolved and inadmissible results within the pool. It then prefers higher minimum-group and overall inclusion, a lower exclusion-rate gap, and more proposed blocks that meet the outcome threshold. Exact ties are resolved by the lexicographic order of the method identifier. Equation 10, not this diagnostic order, selects the station plan. With one fixed template, these inclusion values and the exclusion-rate gap are partly redundant functions of the selected block count. Tied plans receive the same *diagnostic rank*. Lower ranks among plans whose PMF results all fail the fixed pool requirements have no substantive admissibility meaning.

Before generating results with a separate pool for each plan, we froze the hypotheses, code and input hashes, phase log, and criteria for a relevant difference: a change in PMF status or the top-ranked set, any strict pairwise reversal, a rank shift of at least two,

or Kendall $\tau_b < 0.8$. We restrict the ablation to validation; Section 7 states the chronology limits.

5.4 Post hoc sensitivity checks

After inspecting the primary results, we specified two checks based on the five audit metrics. The first expands the set of candidate sites from $|U| = 300$ to 600. It holds $k = 200$, all seven generators, primary time splits, grid, thresholds, and selection rule fixed. The check tests how the high primary $k/|U|$ ratio affects plan diversity and selection.

The Chicago check separates a change in calendar date from a change in hour. It uses the same hour, 07:00–08:00, on three Wednesdays: June 1 for training, June 8 for validation, and June 15 for the test used only for reporting. Seeds 701, 708, and 715 select 2,500 trips from 2,699, 3,578, and 3,727 eligible rows with a Hardship Index. We retain $|U| = 300$ and the validation selection rule. We disable PMF because these checks address candidate-set size and temporal design. Together, the checks test whether the rule’s selection depends on these two design choices. Because we designed the checks post hoc, we do not treat them as confirmatory or use them to estimate sampling uncertainty.

Post hoc bootstrap sensitivity. For the primary 300-site configurations, we additionally draw 10,000 bootstrap samples per city from validation trips, resampling with replacement within each proxy group while retaining its observed size. Every plan uses the same sampled trip indices. In each draw we recompute all group p90 values using linear interpolation and their maximum W_{90} ; the worst group may change. The 2.5th and 97.5th percentiles of the bootstrap distributions summarize W_{90} and paired differences. Plans, group labels and sizes, and the original selection remain fixed; no test trips enter. These exploratory ranges are neither simultaneous nor selection-adjusted.

5.5 Reproducibility

Each output directory stores source hashes; files listing candidate sites and stations in comma-separated values (CSV) format; the fixed grid; validation and test records; the selection hash; proposal pools; and solver results. A phase log with seven recorded events shows that the program records the validation selection before loading the test data. A separately implemented verifier reconstructs candidate sites, audit metrics, selection, PMF pools, block templates, radii, differences between group means, inclusion and exclusion requirements, bounds, and node budgets. It passes 38 checks on the main evaluation artifact. A separate ablation verifier checks the recorded decision order, absence of test inputs, source and plan

hashes, validity of candidate sites, and saved packings. It also checks saved access outcomes, ranks, ties, reversals, and flags for the stated criteria. All 1,570 ablation checks pass. The artifact for the expanded set of candidate sites passes 20 checks, and the Chicago artifact for the same hour across dates passes 11. The bootstrap supplement stores its configuration, seeds, hashes, all draws, and all 21 pairwise differences per city; 11 synthetic tests and 335 separate checks of output arithmetic and input integrity pass. These separately implemented verifiers reproduce the saved computations. The results have not been independently replicated by an external group.

6 Results

6.1 Validation comparison and selection

The validation comparison reports the five audit metrics used in Equation 10. Both settings meet the requirements on the number and trip share of auditable cells, and the audit includes every trip for every plan. Five of seven Porto-SES plans are nondominated. IFkCO uniquely minimizes the first-priority metric, with a worst-group p90 of 885 m, so no lower-priority metric is consulted. Its 97.7% share within the access limit and its cell gap are worse than those of several alternatives. The Pareto label only describes trade-offs.

All seven Chicago-Hardship plans are nondominated. IFkCO has the lowest overall p90 but a worst-group p90 above 11 km, whereas Grid uniquely minimizes the first-priority metric, by 22 m over the next-best plan. The predeclared rule therefore chooses Grid, whose worst-group p90 is 1,735 m. Its 22 m advantage does not establish statistically reliable or practically meaningful superiority over Priority. Overall p90 and worst-group p90 favor different plans.

With validation groups ranging from 159 to 1,952 trips, about 16 observations lie in the smallest group’s upper decile. The reported W_{90} is a tail point estimate and can change with the sample. Table 5 reports the corresponding post hoc sensitivity.

6.2 Test results without reselection

We compare validation and test values for each selected plan; test data never affect selection. Porto-SES overall p90 remains 805 m, while worst-group p90 changes from 885 m to 872 m and the group mean gap narrows from 180 m to 174 m. For Chicago-Hardship, overall p90 changes from 1,442 m to 1,404 m, worst-group p90 changes from 1,735 m to 1,656 m, and the group mean gap narrows from 423 m to 314 m. These are descriptive point estimates. In Chicago, the change in clock time also permits a change in rider composition, so these differences do not isolate temporal generalization from changes in demand.

The Porto-SES and Chicago-Hardship test cell gaps are 161 m and 521 m, down from 319 m and 567 m on validation. Auditable cells contain 98.4%/98.7% (validation/test) of trips on Porto-SES and 95.7%/95.9% on Chicago-Hardship, both above the fixed 90% floor. The respective shares within the access limit are 97.7% at 2 km and 100% at 5 km, unchanged between splits. Cell support affects only the cell gap; every trip enters the other metrics. Because Porto has only eight auditable cells in both splits, its cell gap remains sensitive to individual cells; Chicago has 16 on validation and 15 on test. Together, the validation comparison and frozen test

report answer the research question for the primary audit: the planning authority can trace the choice to one validation evidence set shared by all seven plans, and the selected plan is fixed before the test data are loaded.

6.3 Post hoc sensitivity checks

Table 4 contrasts the constrained primary case with the expanded candidate set: minimum overlap falls and selection changes in both cities. The largest pairwise overlap remains 199 stations in each city: Demand and Group-fair mean in Porto, and Demand and K-medoids in Chicago, for $|U| = 600$. Thus the more diverse configuration does not support a generally preferred generator inferred from the constrained 300-site case. The expanded case was specified after inspecting the primary results and remains exploratory.

In the Chicago check at the same hour on different dates, the rule selects Priority on the June 8 validation data. For this selected plan, the June 15 test gives overall p90 1,195 m, worst-group p90 1,640 m, cell gap 463 m, 100% within the access limit, and 94.0% auditable-cell share over 12 auditable cells. Holding the hour fixed removes the primary design’s time-of-day difference from this check; pairs of plans still share 128–198 stations. These changes in selection show why the authorized candidate sites and the evaluation period are planning inputs rather than implementation details.

Sensitivity to resampled validation trips. Table 5 compares each originally selected plan with the plan having the next-lowest validation W_{90} . Chicago’s range spans both orderings; Porto’s remains below zero for IFkCO minus Grid. This is a pairwise sensitivity result, not a portfolio-wide superiority claim, and does not alter either selection.

6.4 Secondary PMF results, bounds, and pool ablation

For the selected Porto-SES plan on the test data and the block template (3, 1, 1, 7) in display order (very low, low, middle, high), 777 of 1,387 blocks constructed without plan outcomes pass the 250 m threshold. The 10,000-node search returns 174 disjoint blocks with upper bound 184 and a 5.7% gap, reaching the deterministic node limit before the gap closes. The selected blocks cover 41.8% of trips; the lowest group inclusion rate is 34.6%, the exclusion-rate gap is 0.071, and the radius is at most 2.99 km. The result meets all fixed requirements; the reported gap leaves optimality unresolved.

The Chicago-Hardship test pool, template (1, 1, 1, 12), contains 34 blocks; one passes the 500 m threshold. The inclusion floors require 42 blocks overall and 48 for the most restrictive group, already exceeding the single eligible block. After we impose these requirements, HiGHS classifies the model as infeasible and returns no incumbent, so no upper bound, gap, or radius is reported. This result establishes inadmissibility only inside the sparse proposal pool; we do not test PMF feasibility over the complete set of possible blocks. The diagnostic therefore supplies no admissible PMF solution for this Chicago case, rather than evidence of fair access or general PMF infeasibility. The primary plan audit is unchanged.

Table 2: Validation metrics in selection-rule order. Arrows show the preferred direction; bold marks the selected plan and “yes” a nondominated plan. Within L is the share of trips within the endpoint-distance limit.

Setting	Plan generated on training	worst-group p90 (m)↓	overall p90 (m)↓	cell gap (m)↓	within L %↑	group mean gap (m)↓	Pareto?
Porto-SES	Gonzalez	1,404	1,259	222	99.7	203	no
	K-medoids	1,400	1,269	189	99.5	250	no
	Demand	1,239	1,075	101	99.6	222	yes
	Group-fair mean	1,238	1,127	108	99.6	141	yes
	IFkCO	885	805	319	97.7	180	yes
	Priority	1,224	1,107	201	99.7	175	yes
	Grid	1,186	1,145	180	99.6	132	yes
Chicago-Hardship	Gonzalez	1,822	1,508	581	100.0	388	yes
	K-medoids	1,917	1,732	499	100.0	160	yes
	Demand	1,857	1,422	637	100.0	458	yes
	Group-fair mean	1,876	1,508	599	100.0	403	yes
	IFkCO	11,547	1,272	1,894	96.4	3,618	yes
	Priority	1,757	1,347	647	100.0	511	yes
	Grid	1,735	1,442	567	100.0	423	yes

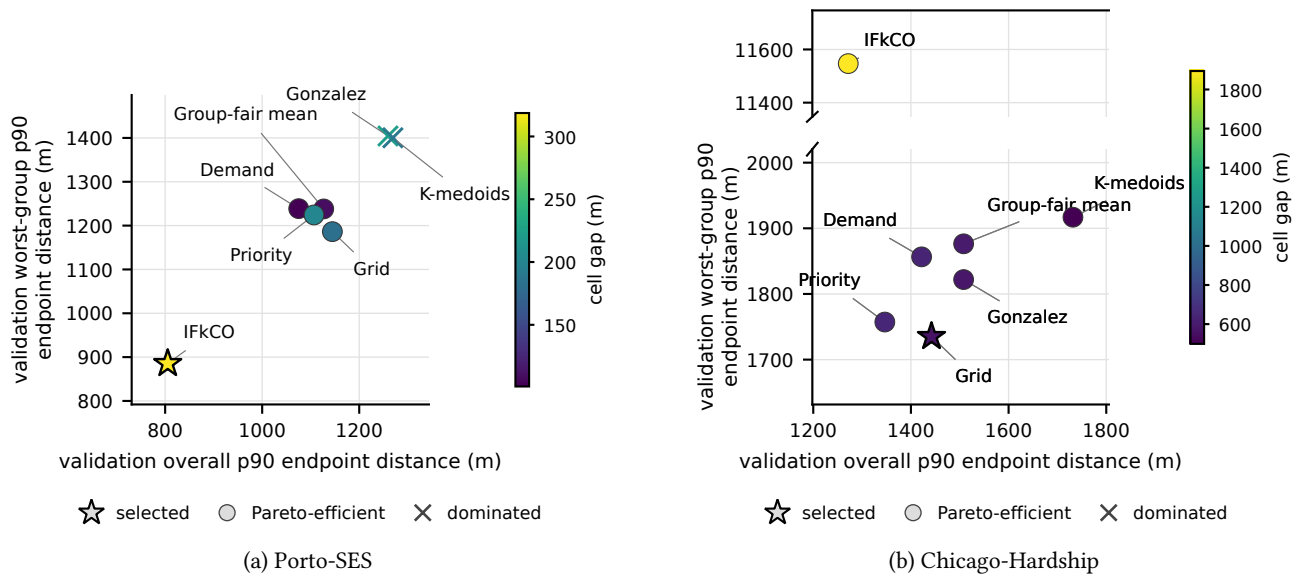


Figure 1: Validation overall versus worst-group p90 endpoint distance. Stars mark selected plans, circles other Pareto plans, and crosses dominated plans (legend below each panel); color encodes the cell gap (colorbar). The Chicago panel’s vertical axis is broken so IFkCO’s outlying worst-group value and the other six plans both stay legible. Table 2 gives the reported values and all five selection metrics.

Table 3: PMF validation ranks under a common pool and one pool per plan (1 = best; tied plans share a rank; * marks the display choice after tie-breaking). Switching from the common pool to one pool per plan reverses Grid and K-medoids in Porto and changes the Chicago ties. All Chicago results are inadmissible; their ranks describe only the diagnostic ordering, not admissible alternatives. The selection rule in Equation 10, not the PMF diagnostic, selects the station plan.

Setting	Pool	IFkCO	Priority	Group-fair mean	Demand	Grid	K-medoids	Gonzalez
Porto-SES	common pool	1*	2	3	4	5	6	7
	pool per plan	1*	2	3	4	6	5	7
Chicago-Hardship	common pool	1	6	4	4	1*	3	6
	pool per plan	2	4	4	4	1*	3	4

Common versus plan-specific pools. Table 3 reports the PMF diagnostic ranks under the order fixed for this ablation. In Porto-SES, the common pool contains 1,168 proposed blocks; the seven pools built separately for each plan contain 291–783. IFkCO has

the unique top diagnostic rank and is the only plan with an admissible PMF result under both designs. The rankings have Kendall $\tau_b = 0.905$ and Spearman $\rho = 0.964$. Grid and K-medoids reverse fifth and sixth place; the maximum shift is one.

Table 4: Candidate-set sensitivity at fixed $k = 200$. Minimum overlap is the smallest number of shared stations across plan pairs. The 300-site cases are primary; the 600-site cases are post hoc. Selection uses validation only.

Setting	$ U $	Min. overlap	Selected generator
Porto-SES	300	113	IFkCO
	600	39	Grid
Chicago-Hardship	300	128	Grid
	600	42	Priority

Table 5: Post hoc paired differences in validation W_{90} (meters). Negative values favor the first plan. Ranges are individual 2.5th–97.5th bootstrap percentiles, without multiple-comparison adjustment.

Setting	Difference	Observed	95% range
Chicago	Grid – Priority	–22	[–111, 229]
Porto	IFkCO – Grid	–301	[–359, –30]

Chicago-Hardship has 16 proposed blocks in the common pool, while its separate pools contain zero to four; every PMF result is inadmissible under the fixed pool requirements in both designs. Five ties change without a strict reversal; the largest rank shift is two, $\tau_b = 0.852$, and $\rho = 0.911$. These ranks do not distinguish admissible alternatives in Chicago.

The ablation shows limited changes in diagnostic ranks across pool designs: Porto retains the same top-ranked admissible PMF result, and Chicago has no admissible result under either design. It nevertheless illustrates why PMF comparisons require a common pool. Although the underlying trip set remains fixed, plan-specific constructors generate different proposal pools, so the packing models optimize over different feasible sets and the resulting PMF values lack a common basis of comparison. For this ablation, the solver certifies all saved pool maxima as optimal within their generated pools, including empty packings. These maxima omit the inclusion and exclusion-rate constraints; admissibility is checked separately.

7 Discussion and Limitations

7.1 Validity controls

The contract makes comparisons traceable to shared observations, reporting cells, and proposal pools. Separating plan generation, validation selection, and test reporting prevents test outcomes from influencing the recorded choice. PMF checks inclusion, locality, and block composition and reports solver bounds; fairness of the station plan remains a policy judgment. The pool constructor changes PMF ranks, while candidate sites and temporal design change plan selection. Each change can be traced to an input declared in the contract.

7.2 Prospective use in urban decision support

For prospective use, a planning authority publishes a versioned contract before receiving proposals.

It specifies the target population and data cutoff, proxy definitions, authorized sites, reporting grid and support rules, access metric and limits, validation period, selection rule, and any proposal pools. Each proposer supplies only a station set and plan hash. An analyst runs the same program for every proposal and publishes validation vectors, Pareto labels, audit validity, PMF admissibility, and hashes; an auditor checks the record without redefining the evidence. The authority owns the normative inputs, decides, may monitor later observations, and starts a new versioned review if an input changes.

An authority can favor a plan by choosing candidate sites, evaluation periods, or metric priorities after inspecting outcomes, even when all plans share the same evidence. A prospective process should publish an externally timestamped contract before comparison and document the reasons for subsequent changes. Hashes alone do not establish this prior commitment.

This is a prospective workflow design; we have not obtained municipal or stakeholder validation for the present data, limits, or selection priorities. Deployment would require authorized sites, access measured on the relevant transport network, policy limits reviewed by stakeholders, repeated time windows, and a public archival record.

7.3 Comparative evaluation for urban decision intelligence

Urban decision intelligence compares plans supplied by classical, learned, or hybrid generators. An evaluation contract provides a shared basis for these comparisons. If a system proposes both an intervention and the units, blocks, or trips used to score it, the contract requires them to remain common (Section 3.1). Our fixed generators isolate this comparison; we make no empirical claim about learned generators.

7.4 Remaining limitations

The protocol records groups, thresholds, and metric priorities but leaves their choice to the planning authority. We test no alternative proxy definitions, thresholds, or priority orders, and estimate neither decision regret nor training variability or uncertainty for learned generators. Porto-SES and Chicago-Hardship use area proxies from years different from the trip data, with unequal sampled group sizes; they describe origin areas, not travelers’ SES [12]. The fixed grid prevents plan-dependent aggregation but remains subject to MAUP [13].

Observed trips represent customers, not full-population station demand, and candidate sites are trip endpoints rather than authorized parcels or stops. Straight-line distance captures geometric proximity, not network access, opportunities, schedules, fares, capacity, safety, or other accessibility dimensions [7]. Some generators lack approximation guarantees; we do not test their original algorithms.

The bootstrap assumes independent, exchangeable trips within proxy groups and conditions on the observed group counts, plans, and windows. It omits spatial or temporal dependence, changing group shares, training uncertainty, and a finite-population correction. We provide no rolling-origin evaluation [21]. Chicago’s 22 m

validation margin relies on a small group's tail, and its primary windows confound split role with clock time. The post hoc same-hour check does not resolve this confound, and expanding candidate sets changes both selections. Stored inputs and code reproduce estimates, not sampling reliability or prospective generalization.

The common PMF pool is a sampled subset of all possible blocks, so even a zero MIP gap would establish optimality only within that pool. Porto-SES has a nonzero gap, while Chicago-Hardship fails the pool requirements. A future clustering hierarchy would need a predeclared rule that preserves the reported radius or diameter limit at the selected cut.

Hashes verify files and phase logs record execution order, but neither provides an external timestamp or public preregistration. The original test was seen before the ablation, sensitivity checks, and bootstrap; the constructor comparison and bootstrap use validation only, and added analyses are labeled post hoc. The package has no archival DOI at the time of writing.

8 Conclusion

A frozen evaluation contract attributes score differences to plan outcomes within one declared comparison. Both retrospective cases show that changing candidate sites or temporal design can change the selected plan. The protocol exposes this dependence before decisions. It makes populations, sites, reporting cells, thresholds, and metric priorities explicit, while PMF conclusions remain limited to sampled pools. Authorities should publish the contract and restart evaluation whenever an input changes.

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